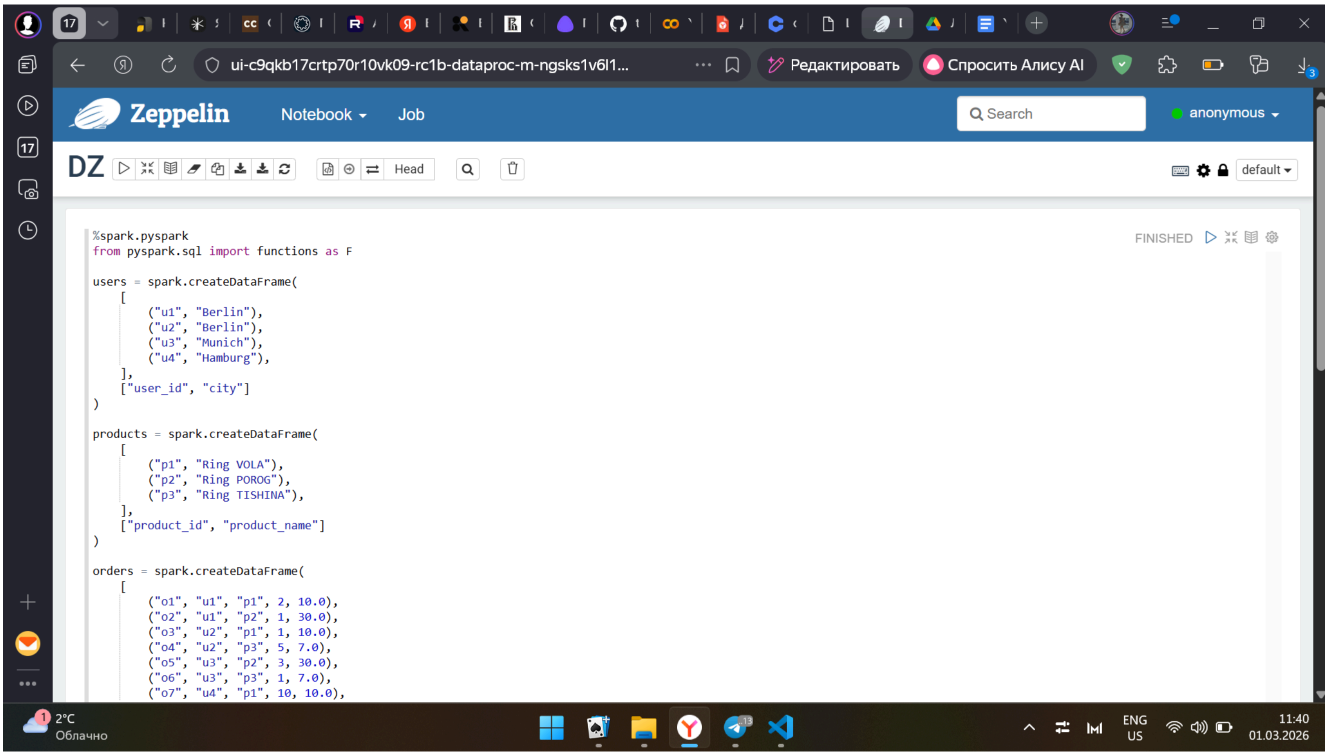
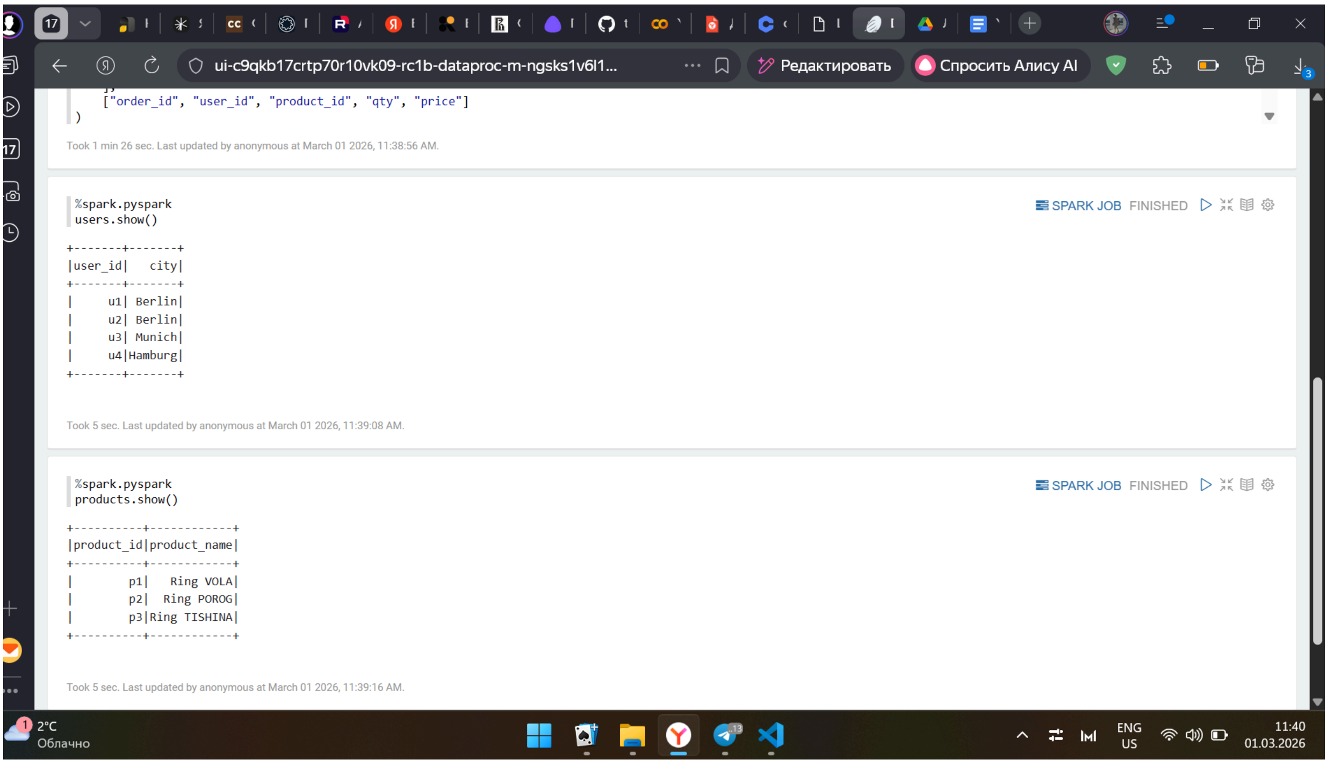
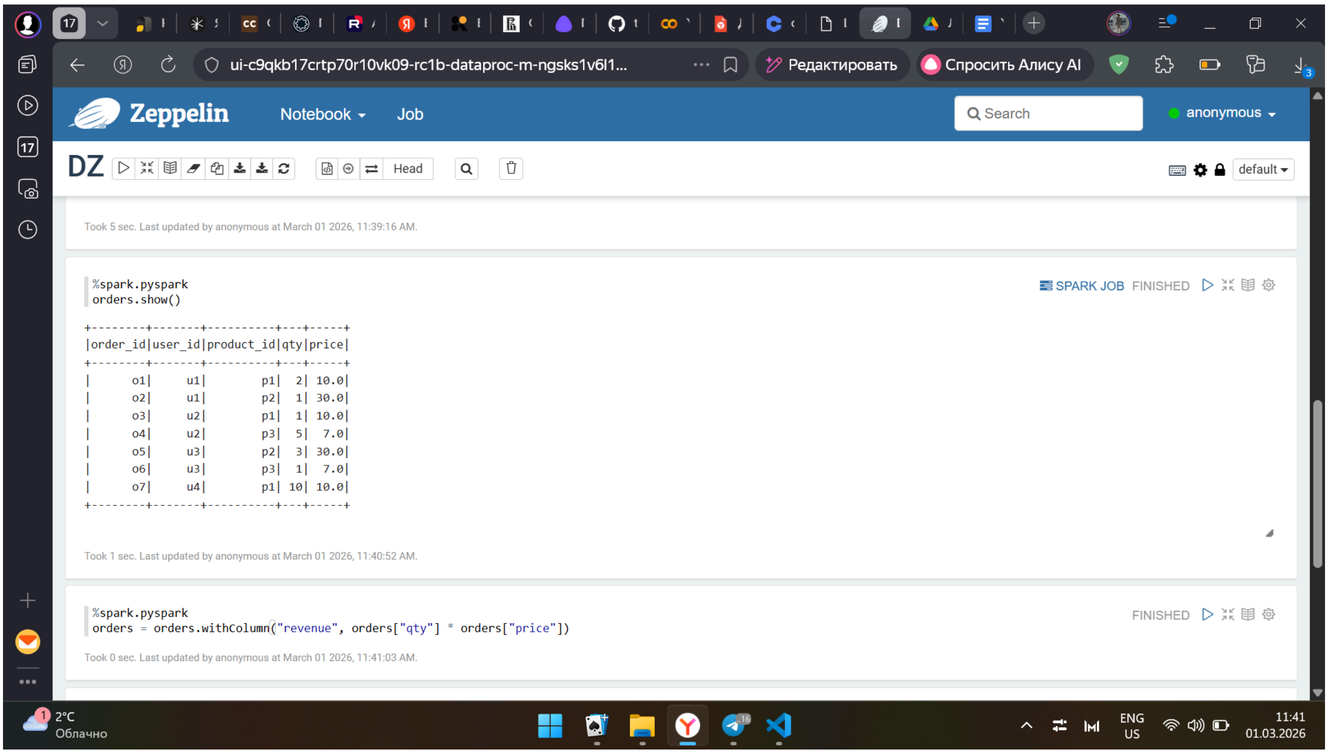
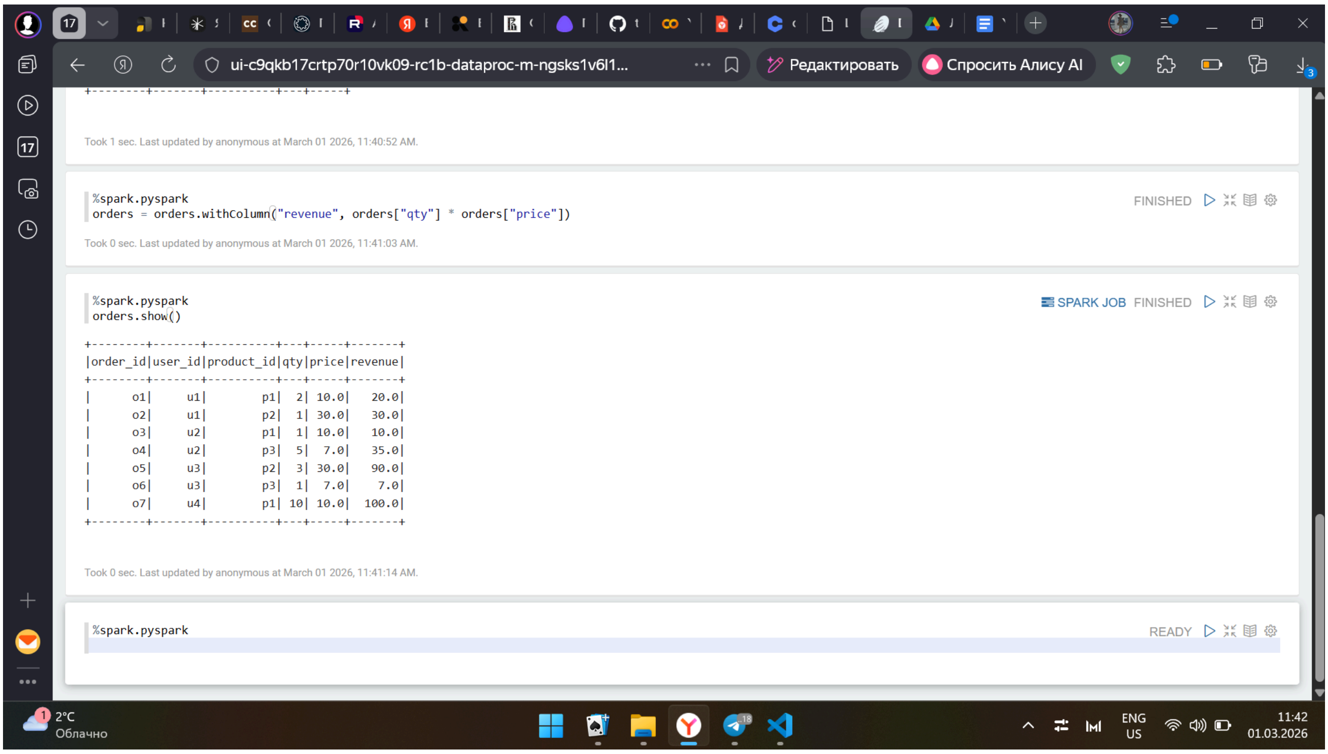
Загрузка и вывод данных



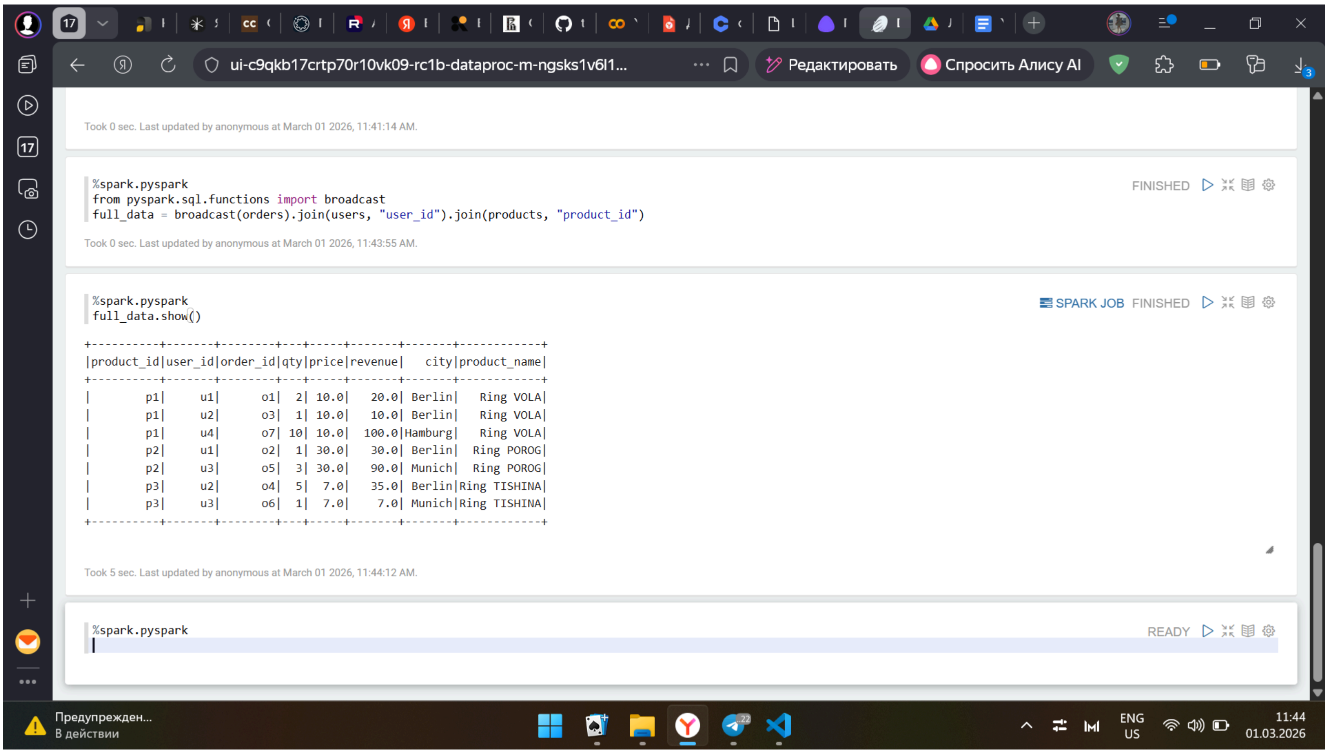



ui-c9qkb17crtp70r10vk09-rc1b-dataproc-m-ngsks1v6l1...
Редактировать
Спросить Алису AI
Zeppelin
Notebook
Job
Search
anonymous
DZ
Head
default
Took 5 sec. Last updated by anonymous at March 01 2026, 11:39:16 AM.
%spark.pyspark
orders.show()
SPARK JOB FINISHED
+--------+-------+----------+---+-----+
|order_id|user_id|product_id|qty|price|
+--------+-------+----------+---+-----+
o1	u1	p1	2	10.0
o2	u1	p2	1	30.0
o3	u2	p1	1	10.0
o4	u2	p3	5	7.0
o5	u3	p2	3	30.0
o6	u3	p3	1	7.0
o7	u4	p1	10	10.0
+--------+-------+----------+---+-----+
Took 1 sec. Last updated by anonymous at March 01 2026, 11:40:52 AM.
%spark.pyspark
orders = orders.withColumn("revenue", orders["qty"] * orders["price"])
FINISHED
Took 0 sec. Last updated by anonymous at March 01 2026, 11:41:03 AM.
2°C
Облачно
ENG
US
11:41
01.03.2026

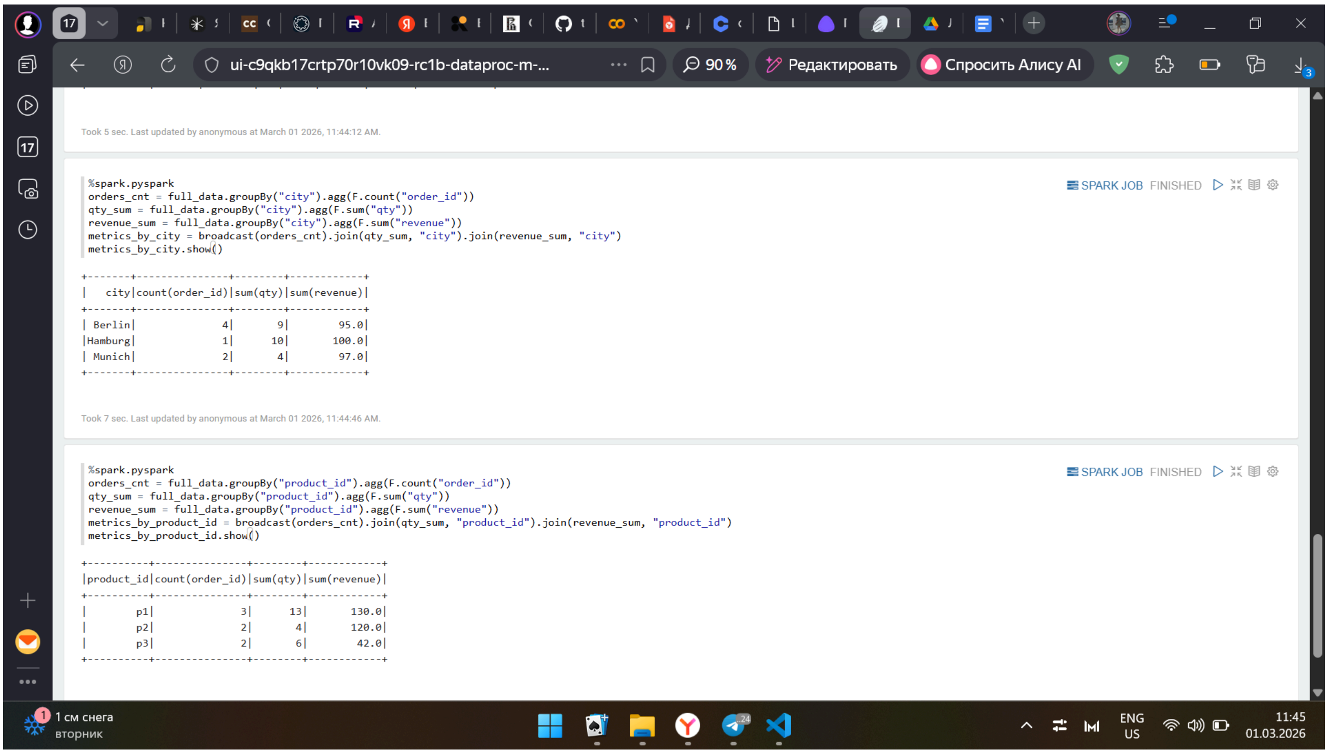
Задание 1


ui-c9qkb17crtp70r10vk09-rc1b-dataproc-m-ngsks1v6l1...
Редактировать
Спросить Алису AI
Took 1 sec. Last updated by anonymous at March 01 2026, 11:40:52 AM.
%spark.pyspark
orders = orders.withColumn("revenue", orders["qty"] * orders["price"])
FINISHED
Took 0 sec. Last updated by anonymous at March 01 2026, 11:41:03 AM.
%spark.pyspark
orders.show()
SPARK JOB FINISHED
+--------+-------+----------+---+-----+-------+
|order_id|user_id|product_id|qty|price|revenue|
+--------+-------+----------+---+-----+-------+
o1	u1	p1	2	10.0	20.0
o2	u1	p2	1	30.0	30.0
o3	u2	p1	1	10.0	10.0
o4	u2	p3	5	7.0	35.0
o5	u3	p2	3	30.0	90.0
o6	u3	p3	1	7.0	7.0
o7	u4	p1	10	10.0	100.0
+--------+-------+----------+---+-----+-------+
Took 0 sec. Last updated by anonymous at March 01 2026, 11:41:14 AM.
%spark.pyspark
READY
2°C
Облачно
ENG
US
11:42
01.03.2026

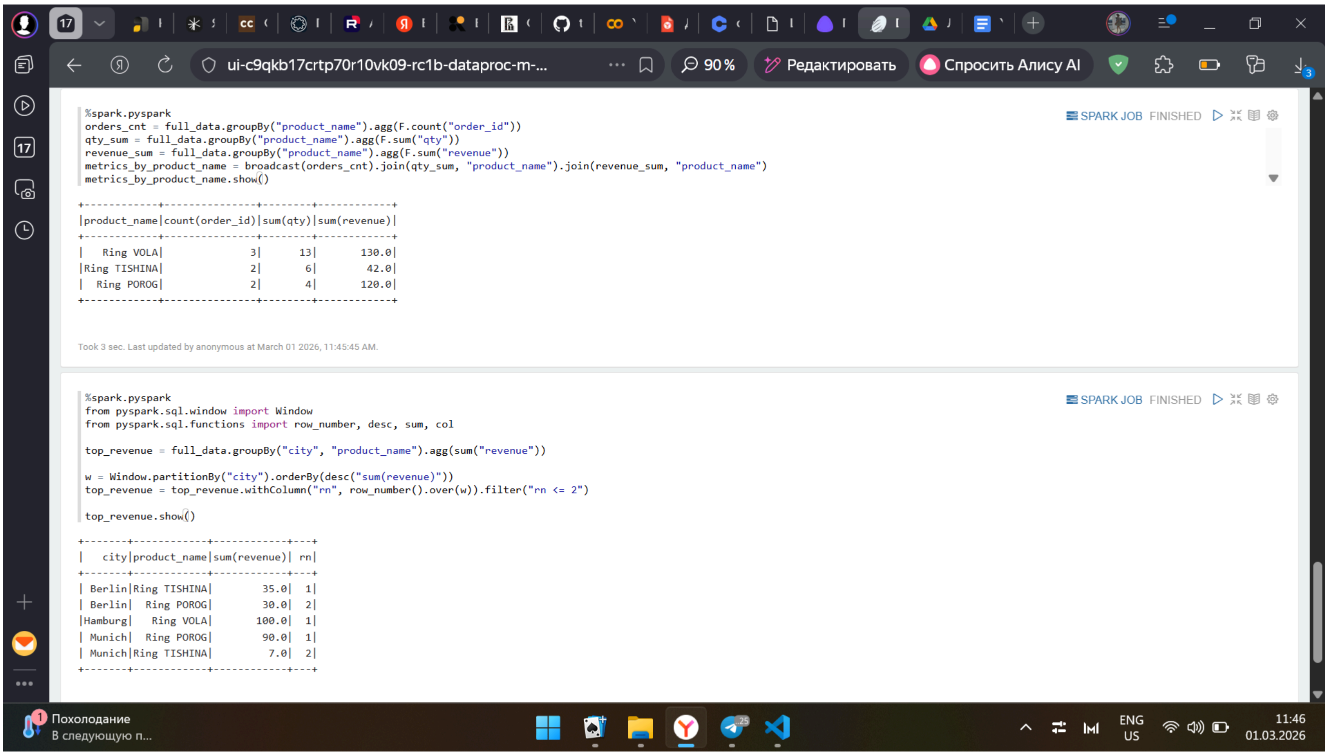
Задание 2



Задание 3



Задание 4

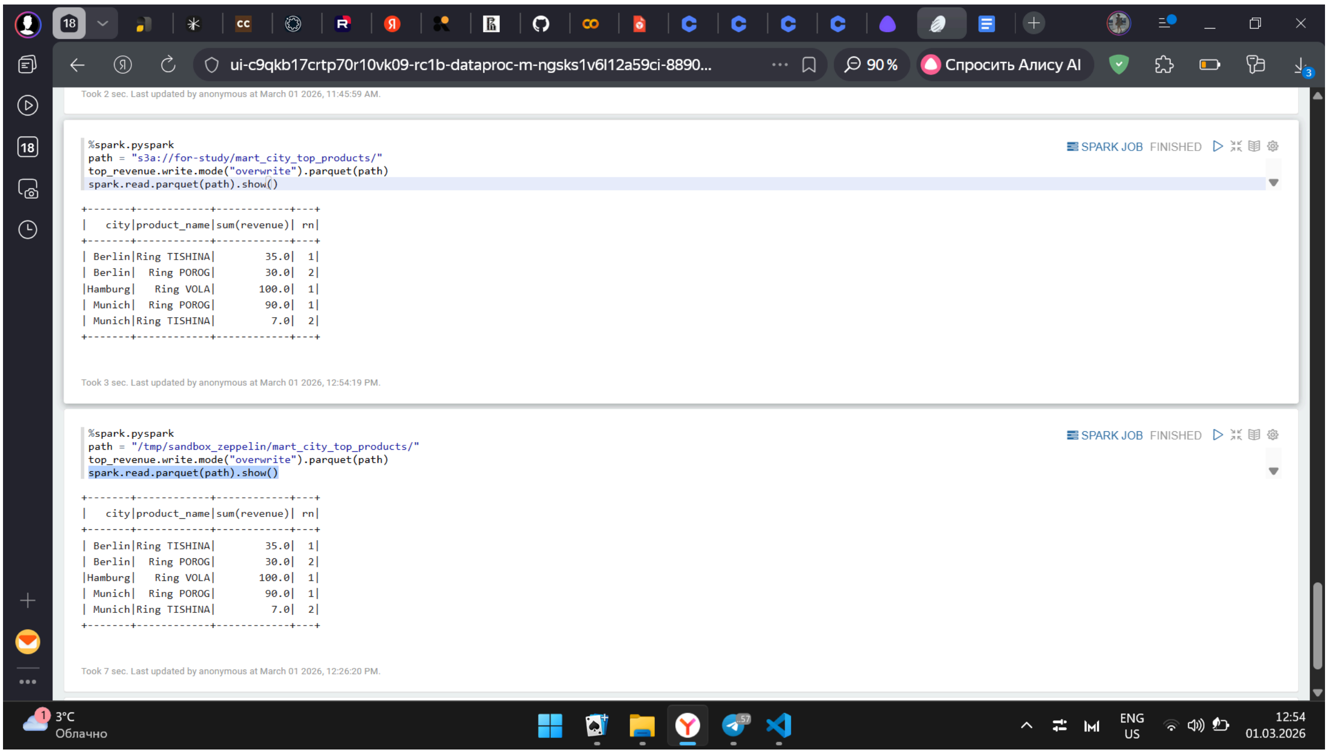



```
%spark.pyspark
orders_cnt = full_data.groupBy("product_name").agg(F.count("order_id"))
qty_sum = full_data.groupBy("product_name").agg(F.sum("qty"))
revenue_sum = full_data.groupBy("product_name").agg(F.sum("revenue"))
metrics_by_product_name = broadcast(orders_cnt).join(qty_sum, "product_name").join(revenue_sum, "product_name")
metrics_by_product_name.show()

+------------+--------------+--------+------------+
|product_name|count(order_id)|sum(qty)|sum(revenue)|
+------------+--------------+--------+------------+
|   Ring VOLA|             3|      13|       130.0|
|Ring TISHINA|             2|       6|        42.0|
|  Ring POROG|             2|       4|       120.0|
+------------+--------------+--------+------------+
```

Took 3 sec. Last updated by anonymous at March 01 2026, 11:45:45 AM.


```
%spark.pyspark
from pyspark.sql.window import Window
from pyspark.sql.functions import row_number, desc, sum, col

top_revenue = full_data.groupBy("city", "product_name").agg(sum("revenue"))

w = Window.partitionBy("city").orderBy(desc("sum(revenue)"))
top_revenue = top_revenue.withColumn("rn", row_number().over(w)).filter("rn <= 2")

top_revenue.show()

+-------+------------+------------+---+
|   city|product_name|sum(revenue)| rn|
+-------+------------+------------+---+
| Berlin|Ring TISHINA|        35.0|  1|
| Berlin|  Ring POROG|        30.0|  2|
|Hamburg|   Ring VOLA|       100.0|  1|
| Munich|  Ring POROG|        90.0|  1|
| Munich|Ring TISHINA|         7.0|  2|
+-------+------------+------------+---+
```


Задание 5 и 6



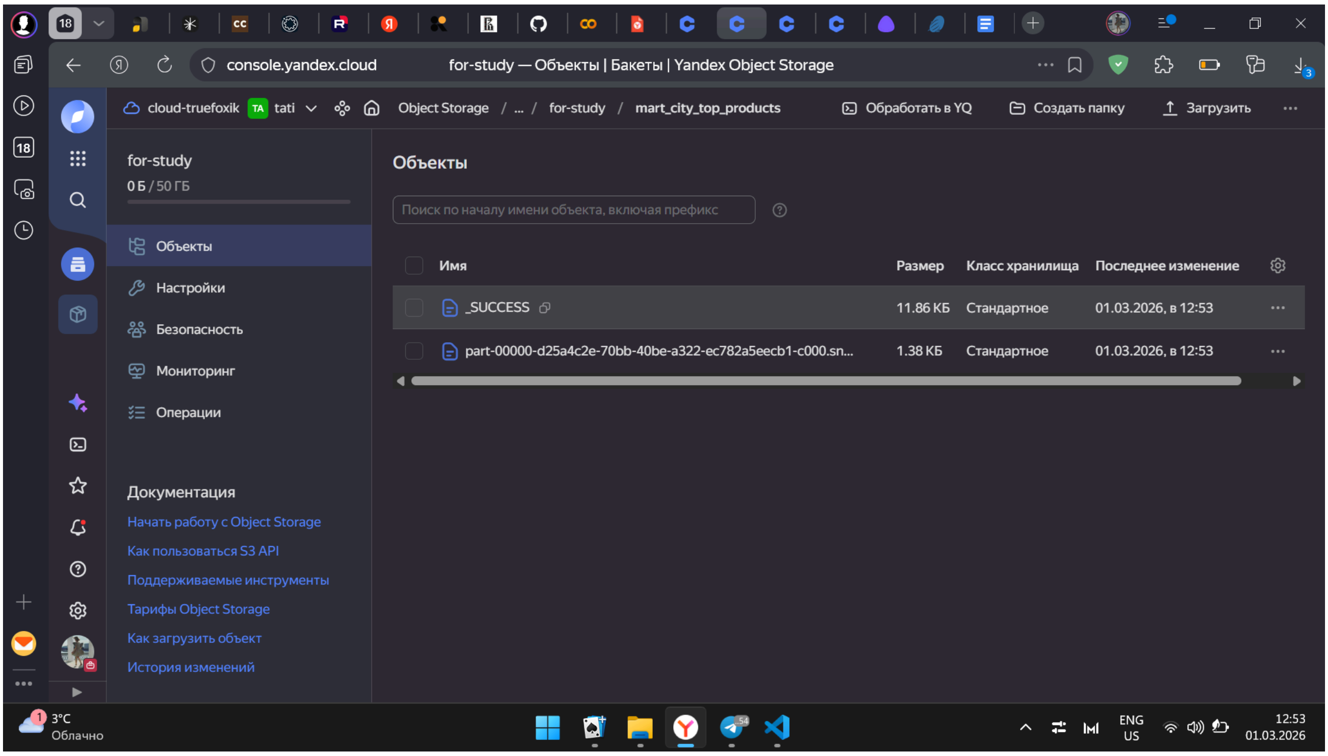
Took 2 sec. Last updated by anonymous at March 01 2026, 11:45:59 AM.


```
%spark.pyspark
path = "s3a://for-study/mart_city_top_products/"
top_revenue.write.mode("overwrite").parquet(path)
spark.read.parquet(path).show()

+-------+------------+------------+---+
|   city|product_name|sum(revenue)| rn|
+-------+------------+------------+---+
| Berlin|Ring TISHINA|        35.0|  1|
| Berlin|  Ring POROG|        30.0|  2|
|Hamburg|   Ring VOLA|       100.0|  1|
| Munich|  Ring POROG|        90.0|  1|
| Munich|Ring TISHINA|         7.0|  2|
+-------+------------+------------+---+
```

Took 3 sec. Last updated by anonymous at March 01 2026, 12:54:19 PM.


```
%spark.pyspark
path = "/tmp/sandbox_zeppelin/mart_city_top_products/"
top_revenue.write.mode("overwrite").parquet(path)
spark.read.parquet(path).show()

+-------+------------+------------+---+
|   city|product_name|sum(revenue)| rn|
+-------+------------+------------+---+
| Berlin|Ring TISHINA|        35.0|  1|
| Berlin|  Ring POROG|        30.0|  2|
|Hamburg|   Ring VOLA|       100.0|  1|
| Munich|  Ring POROG|        90.0|  1|
| Munich|Ring TISHINA|         7.0|  2|
+-------+------------+------------+---+
```

Took 7 sec. Last updated by anonymous at March 01 2026, 12:26:20 PM.


console.yandex.cloud
for-study — Объекты | Бакеты | Yandex Object Storage
cloud-truefoxik
tati
Object Storage / ... / for-study / mart_city_top_products
Обработать в YQ
Создать папку
Загрузить
for-study
0 Б / 50 ГБ
Объекты
Настройки
Безопасность
Мониторинг
Операции
Документация
Начать работу с Object Storage
Как пользоваться S3 API
Поддерживаемые инструменты
Тарифы Object Storage
Как загрузить объект
История изменений
Объекты
Поиск по началу имени объекта, включая префикс
Имя
Размер
Класс хранилища
Последнее изменение
_SUCCESS
11.86 КБ
Стандартное
01.03.2026, в 12:53
part-00000-d25a4c2e-70bb-40be-a322-ec782a5eecb1-c000.sn...
1.38 КБ
Стандартное
01.03.2026, в 12:53
3°C
Облачно
ENG
US
12:53
01.03.2026